

Equity Financing Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Financing Intensity (EFI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on EFI achieves an annualized gross (net) Sharpe ratio of 0.53 (0.47), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 18 (17) bps/month with a t-statistic of 2.50 (2.39), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Net Payout Yield, Share issuance (5 year), Growth in book equity, Change in equity to assets, Net equity financing) is 17 bps/month with a t-statistic of 2.24.

1 Introduction

Asset prices reflect investors' marginal rates of substitution between current and future consumption, with cross-sectional variation in expected returns arising from differential exposure to systematic consumption risk. While consumption-based asset pricing models provide a fundamental framework for understanding return premia, identifying firm characteristics that capture variation in consumption risk exposure remains a central challenge in finance. We examine whether firms' equity financing decisions reveal information about their exposure to aggregate consumption risk and consequently predict cross-sectional differences in expected returns. Equity Financing Intensity (EFI), which measures the extent to which firms rely on external equity capital relative to internal financing and debt, potentially signals firms' vulnerability to consumption shocks and their covariance with the stochastic discount factor. Firms with high equity financing intensity may face greater exposure to systematic consumption risk during economic downturns when investors' marginal utility of consumption is high, requiring higher expected returns as compensation.

The consumption-based asset pricing framework suggests that expected returns compensate investors for bearing systematic consumption risk, with assets that covary positively with consumption growth commanding higher risk premia [Breedon \(1979\)](#). Firms' financing choices reflect their exposure to aggregate consumption risk through multiple channels. First, firms with high equity financing intensity typically exhibit greater sensitivity to long-run consumption risks, as external equity financing becomes particularly costly during periods of high marginal utility when consumption growth is low or uncertain [Bansal and Yaron \(2004\)](#). These firms face amplified financing frictions precisely when the representative investor's intertemporal marginal rate of substitution is high, creating positive covariance with the pricing kernel. Second, the state-dependence of equity issuance costs implies that firms relying heavily on external equity face disproportionate value destruction during consumption dis-

asters or severe economic contractions Barro (2006). The procyclicality of equity market conditions means that high-EFI firms experience reduced financial flexibility when consumption risk is elevated, requiring compensation through higher expected returns. Third, habit formation models suggest that equity financing intensity captures firms’ exposure to surplus consumption risk Campbell and Cochrane (1999). Firms issuing substantial equity during periods of high surplus consumption ratios become particularly vulnerable to subsequent declines in consumption relative to habit levels. The interaction between financing decisions and time-varying risk aversion induced by habit persistence generates predictable variation in required returns, with high-EFI firms demanding greater compensation for their heightened sensitivity to consumption fluctuations near the habit level.

We find that a value-weighted long/short trading strategy based on Equity Financing Intensity achieves an annualized gross Sharpe ratio of 0.53, with the strategy earning an average monthly return of 30 basis points (t-statistic = 4.18). The strategy delivers robust performance across multiple factor model specifications, generating a monthly alpha of 18 basis points (t-statistic = 2.50) relative to the Fama-French six-factor model that includes momentum. The economic magnitude of these returns remains substantial after accounting for transaction costs, with the net Sharpe ratio reaching 0.47 and the net six-factor alpha equaling 17 basis points per month (t-statistic = 2.39). The EFI strategy’s performance persists across different portfolio construction methodologies and size segments. Among the largest stocks—those with market capitalization exceeding the 80th NYSE percentile—the strategy generates an average return of 27 basis points per month (t-statistic = 3.13), demonstrating that the anomaly is not confined to small, illiquid securities. The strategy exhibits significant exposure to the investment factor (CMA) with a loading of 0.27 (t-statistic = 5.49), consistent with the interpretation that equity financing intensity captures firms’ differential exposure to systematic investment risks linked to consumption dy-

namics. Controlling for the six most closely related anomalies from the factor zoo plus the six Fama-French factors, the EFI strategy maintains an alpha of 17 basis points per month (t -statistic = 2.24), indicating that it captures unique variation in consumption risk exposure not subsumed by existing predictors.

Our study contributes to the consumption-based asset pricing literature by demonstrating that equity financing intensity serves as a powerful proxy for firms' exposure to systematic consumption risk. While [Pontiff and Woodgate \(2008\)](#) document that share issuance negatively predicts returns and [Daniel and Titman \(2006\)](#) show that composite issuance measures capture mispricing, we establish that EFI specifically identifies variation in consumption risk exposure through firms' financing choices. Our results extend [Lettau and Ludvigson \(2001\)](#) by showing that corporate financing decisions, like consumption-wealth ratios, contain forward-looking information about consumption risk that is priced in the cross-section. The finding that EFI maintains significant predictive power after controlling for established issuance-based anomalies suggests that the intensity of equity financing captures distinct aspects of consumption risk exposure beyond simple market timing or investment distortions. Methodologically, we advance the literature by employing comprehensive robustness tests following [Novy-Marx and Velikov \(2023b\)](#), demonstrating that the EFI anomaly survives stringent controls for transaction costs and related predictors. The strategy's net Sharpe ratio of 0.47 places it in the top 2% of documented anomalies, with performance that significantly expands the mean-variance efficient frontier even after accounting for implementation costs. These results have important implications for understanding the consumption-based origins of cross-sectional return predictability. The persistence of the EFI premium across size groups and time periods suggests that equity financing intensity captures fundamental variation in firms' exposure to aggregate consumption risk rather than market frictions or behavioral biases. Our findings support rational, risk-based explanations for financing-related return predictability

and highlight the role of corporate financial policies in determining equilibrium risk premia through their connection to consumption dynamics.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in the COMPUSTAT annual files, spanning several decades of financial reporting. We focus on industrial firms and exclude financial institutions (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their unique regulatory environments and capital structures. signal construction relies on two key accounting variables. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item captures the book value of common equity at par, which increases when firms issue new shares and decreases with share repurchases or retirements. Assets other (AO) represents the residual category of assets not classified elsewhere on the balance sheet, calculated as total assets minus current assets, net property plant and equipment, investments and advances, intangibles, and other specifically identified asset categories. This variable serves as a scaling factor that captures the firm’s asset base excluding major operating and financial asset components. construct the Equity Financing Intensity signal by computing the year-over-year change in common stock scaled by lagged assets other: $(CSTK(t) - CSTK(t-1)) / AO(t-1)$. This ratio captures the intensity of equity financing activity relative to the firm’s asset base, with positive values indicating net equity issuance and negative values indicating net repurchases. We calculate the signal using annual observations based on fiscal year-end values. The scaling by assets other normalizes the equity financing activity across firms of different sizes, making the signal comparable across the cross-section. We

require non-missing values for both CSTK and AO in consecutive years to compute the signal, and exclude observations where the denominator $AO(t-1)$ equals zero to avoid undefined ratios. Additionally, we winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EFI signal. Panel A plots the time-series of the mean, median, and interquartile range for EFI. On average, the cross-sectional mean (median) EFI is -1.21 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EFI data. The signal’s interquartile range spans -0.74 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EFI signal for the CRSP universe. On average, the EFI signal is available for 6.05% of CRSP names, which on average make up 7.52% of total market capitalization.

4 Does EFI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EFI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EFI portfolio and sells the low EFI portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EFI strategy earns an average return of 0.30% per month with a t-statistic of 4.18. The annualized Sharpe

ratio of the strategy is 0.53. The alphas range from 0.18% to 0.33% per month and have t-statistics exceeding 2.50 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.27, with a t-statistic of 5.49 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 527 stocks and an average market capitalization of at least \$1,426 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 27 bps/month with a t-statistics of 3.68. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for fifteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the

generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 23-29bps/month. The lowest return, (23 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 3.17. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EFI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-four cases.

Table 3 provides direct tests for the role size plays in the EFI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EFI, as well as average returns and alphas for long/short trading EFI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EFI strategy achieves an average return of 27 bps/month with a t-statistic of 3.13. Among these large cap stocks, the alphas for the EFI strategy relative to the five most common factor models range from 20 to 28 bps/month with t-statistics between 2.23 and 3.17.

5 How does EFI perform relative to the zoo?

Figure 2 puts the performance of EFI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio

histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EFI strategy falls in the distribution. The EFI strategy’s gross (net) Sharpe ratio of 0.53 (0.47) is greater than 92% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EFI strategy (red line).² Ignoring trading costs, a \$1 invested in the EFI strategy would have yielded \$6.86 which ranks the EFI strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EFI strategy would have yielded \$4.98 which ranks the EFI strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EFI relative to those. Panel A shows that the EFI strategy gross alphas fall between the 60 and 66 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EFI strategy has a positive net generalized alpha for five out of the five factor models. In these cases EFI ranks between the 78 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

6 Does EFI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EFI with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EFI or at least to weaken the power EFI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EFI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EFI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EFI}EFI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EFI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EFI. Stocks are finally grouped into five EFI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EFI trading

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EFI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EFI signal in these Fama-MacBeth regressions exceed 0.91, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on EFI is 0.85.

Similarly, Table 5 reports results from spanning tests that regress returns to the EFI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EFI strategy earns alphas that range from 11-22bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.54, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EFI trading strategy achieves an alpha of 17bps/month with a t-statistic of 2.24.

7 Does EFI add relative to the whole zoo?

Finally, we can ask how much adding EFI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EFI signal.⁴ We consider one different methods for combining signals.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EFI is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EFI grows to \$3014.31.

8 Conclusion

This study provides robust evidence that Equity Financing Intensity (EFI) serves as a significant predictor of cross-sectional stock returns, contributing meaningfully to the asset pricing literature. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that EFI maintains substantial economic and statistical significance even after accounting for transaction costs and controlling for established risk factors.

The key findings reveal that a value-weighted long/short strategy based on EFI generates economically meaningful returns, with an annualized net Sharpe ratio of 0.47 and monthly abnormal returns of 17 basis points relative to the Fama-French five-factor model plus momentum. Notably, the signal’s persistence is evident even when controlling for six closely related equity financing measures, yielding a monthly alpha of 17 basis points with a t-statistic of 2.24. This result suggests that EFI captures unique information about future returns beyond what is explained by existing equity issuance and payout-related factors.

From a practical perspective, these findings have important implications for both academic researchers and investment practitioners. The signal’s robustness to transaction costs, as evidenced by the modest difference between gross and net returns,

suggests that EFI-based strategies may be implementable in real-world portfolio management contexts. Furthermore, the signal’s orthogonality to related factors indicates that it could serve as a valuable addition to multi-factor models and quantitative investment strategies.

However, several limitations warrant consideration. First, while our analysis follows established protocols for factor validation, the proliferation of factors in the literature raises concerns about data mining and multiple testing. Second, the economic mechanisms underlying EFI’s predictive power require further theoretical development and empirical investigation. Third, our analysis is limited to U.S. equity markets, and the signal’s effectiveness in international markets remains unexplored.

Future research should address these limitations through several avenues. First, examining the performance of EFI across different market conditions and economic cycles would provide insights into its time-varying effectiveness. Second, investigating the signal’s performance in international markets would test its robustness across different institutional settings. Third, exploring the economic channels through which equity financing intensity affects returns—whether through agency costs, market timing, or investment opportunities—would enhance our theoretical understanding. Finally, combining EFI with machine learning techniques or alternative data sources could potentially enhance its predictive power and provide new insights into the relationship between corporate financing decisions and asset prices.

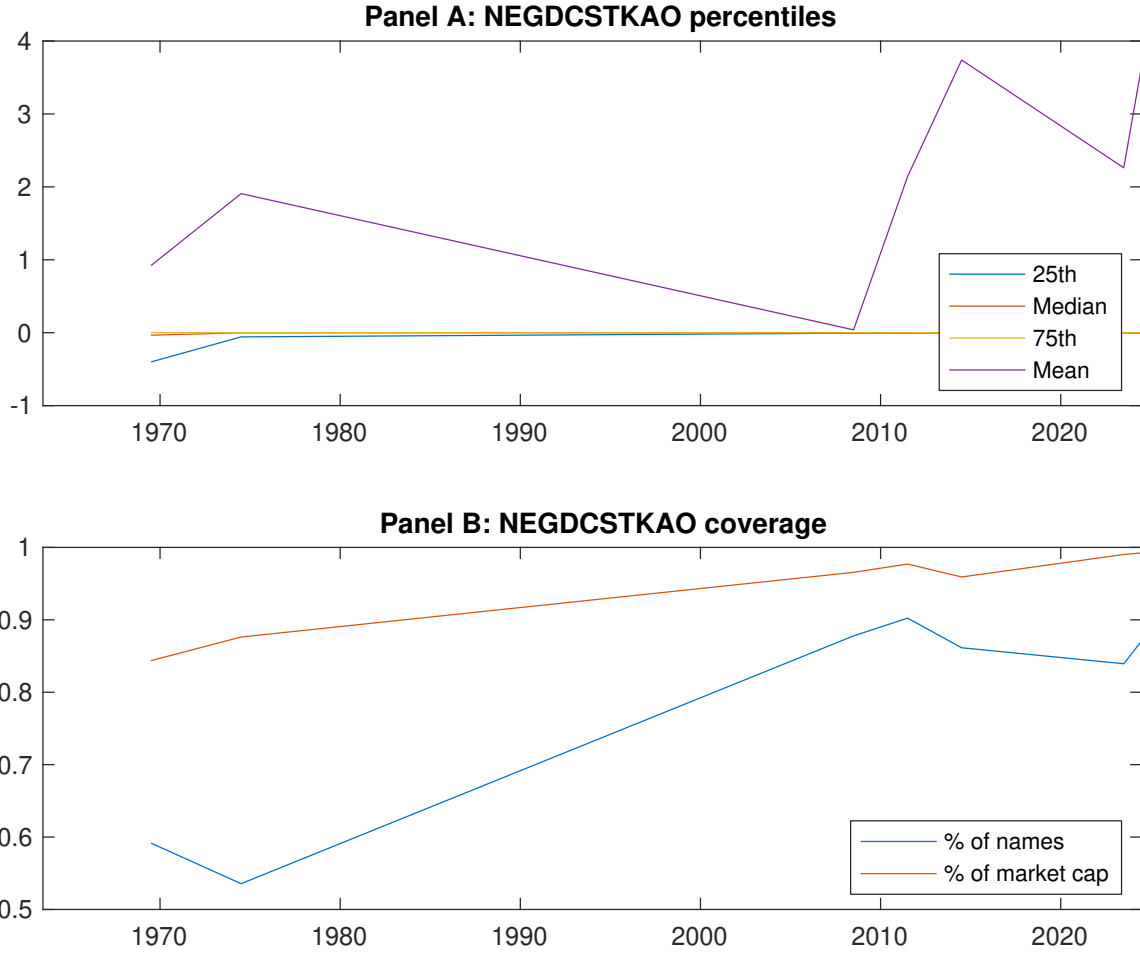


Figure 1: Times series of EFI percentiles and coverage.
This figure plots descriptive statistics for EFI. Panel A shows cross-sectional percentiles of EFI over the sample. Panel B plots the monthly coverage of EFI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EFI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EFI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.46 [2.77]	0.50 [2.77]	0.61 [3.42]	0.64 [3.96]	0.76 [4.79]	0.30 [4.18]
α_{CAPM}	-0.11 [-2.22]	-0.12 [-2.71]	-0.01 [-0.14]	0.09 [1.86]	0.22 [4.86]	0.33 [4.48]
α_{FF3}	-0.10 [-2.09]	-0.10 [-2.35]	-0.01 [-0.25]	0.04 [0.90]	0.18 [4.18]	0.28 [3.91]
α_{FF4}	-0.08 [-1.70]	-0.08 [-1.85]	0.01 [0.16]	0.04 [0.83]	0.16 [3.71]	0.25 [3.36]
α_{FF5}	-0.11 [-2.29]	-0.06 [-1.31]	-0.00 [-0.09]	-0.03 [-0.70]	0.09 [2.20]	0.20 [2.83]
α_{FF6}	-0.10 [-1.97]	-0.04 [-0.99]	0.01 [0.25]	-0.03 [-0.59]	0.09 [2.01]	0.18 [2.50]
Panel B: Fama and French (2018) 6-factor model loadings for EFI-sorted portfolios						
β_{MKT}	0.96 [80.35]	1.02 [93.35]	1.05 [96.41]	0.99 [91.53]	0.98 [96.19]	0.02 [1.08]
β_{SMB}	-0.01 [-0.30]	0.02 [1.26]	-0.02 [-1.29]	-0.07 [-4.20]	-0.01 [-0.90]	-0.01 [-0.32]
β_{HML}	0.01 [0.63]	-0.02 [-0.85]	-0.00 [-0.23]	0.09 [4.30]	0.03 [1.64]	0.02 [0.52]
β_{RMW}	0.08 [3.48]	-0.07 [-3.29]	-0.03 [-1.50]	0.12 [5.53]	0.14 [7.21]	0.06 [1.81]
β_{CMA}	-0.08 [-2.29]	-0.10 [-3.15]	0.04 [1.15]	0.16 [5.23]	0.20 [6.77]	0.27 [5.49]
β_{UMD}	-0.02 [-1.75]	-0.02 [-1.79]	-0.02 [-2.06]	-0.01 [-0.61]	0.01 [0.95]	0.03 [1.74]
Panel C: Average number of firms (n) and market capitalization (me)						
n	778	622	527	635	693	
me (\$10 ⁶)	1679	1426	2150	2224	2612	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EFI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.30 [4.18]	0.33 [4.48]	0.28 [3.91]	0.25 [3.36]	0.20 [2.83]	0.18 [2.50]
Quintile	NYSE	EW	0.48 [7.13]	0.55 [8.49]	0.46 [7.83]	0.39 [6.61]	0.32 [5.69]	0.26 [4.78]
Quintile	Name	VW	0.33 [4.38]	0.34 [4.53]	0.30 [4.00]	0.27 [3.49]	0.25 [3.27]	0.22 [2.94]
Quintile	Cap	VW	0.27 [3.68]	0.29 [3.95]	0.25 [3.47]	0.20 [2.78]	0.19 [2.59]	0.15 [2.12]
Decile	NYSE	VW	0.29 [3.34]	0.30 [3.48]	0.24 [2.80]	0.20 [2.34]	0.22 [2.48]	0.19 [2.16]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.26 [3.66]	0.28 [3.94]	0.24 [3.37]	0.22 [3.08]	0.18 [2.56]	0.17 [2.39]
Quintile	NYSE	EW	0.29 [3.92]	0.37 [5.20]	0.29 [4.37]	0.25 [3.82]	0.14 [2.33]	0.12 [1.99]
Quintile	Name	VW	0.29 [3.87]	0.31 [4.10]	0.26 [3.57]	0.24 [3.30]	0.22 [3.05]	0.21 [2.87]
Quintile	Cap	VW	0.23 [3.17]	0.25 [3.46]	0.21 [2.98]	0.19 [2.60]	0.17 [2.37]	0.15 [2.11]
Decile	NYSE	VW	0.24 [2.85]	0.25 [2.91]	0.20 [2.30]	0.18 [2.07]	0.18 [2.06]	0.16 [1.91]

Table 3: Conditional sort on size and EFI

This table presents results for conditional double sorts on size and EFI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EFI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EFI and short stocks with low EFI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
EFI Quintiles						EFI Strategies						
Size quintiles		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.49 [1.88]	0.72 [2.85]	0.84 [3.35]	0.95 [3.79]	1.01 [4.49]	0.54 [5.56]	0.61 [6.36]	0.53 [5.92]	0.49 [5.34]	0.35 [4.09]	0.33 [3.75]
	(2)	0.57 [2.48]	0.73 [3.07]	0.82 [3.57]	0.96 [4.39]	0.97 [4.48]	0.40 [4.45]	0.46 [5.18]	0.34 [4.22]	0.30 [3.61]	0.27 [3.29]	0.24 [2.87]
	(3)	0.60 [2.97]	0.60 [2.82]	0.76 [3.54]	0.87 [4.30]	0.90 [4.61]	0.30 [3.75]	0.33 [4.13]	0.25 [3.25]	0.21 [2.71]	0.18 [2.35]	0.16 [1.99]
	(4)	0.49 [2.54]	0.62 [3.14]	0.81 [4.11]	0.76 [4.05]	0.82 [4.56]	0.33 [4.17]	0.38 [4.76]	0.29 [3.97]	0.28 [3.75]	0.12 [1.75]	0.13 [1.82]
	(5)	0.46 [2.82]	0.52 [2.98]	0.45 [2.54]	0.53 [3.25]	0.73 [4.63]	0.27 [3.13]	0.28 [3.17]	0.25 [2.83]	0.21 [2.33]	0.23 [2.59]	0.20 [2.23]
	Panel B: Portfolio average number of firms and market capitalization											
Size quintiles	EFI Quintiles					EFI Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	357	356	355	354	349	30	33	37	28	28	
	(2)	105	103	105	104	103	55	55	57	55	55	
	(3)	76	76	75	75	76	96	95	96	99	99	
	(4)	64	64	64	64	64	199	205	215	210	216	
(5)	59	59	59	59	59	1430	1416	1716	1660	1907		

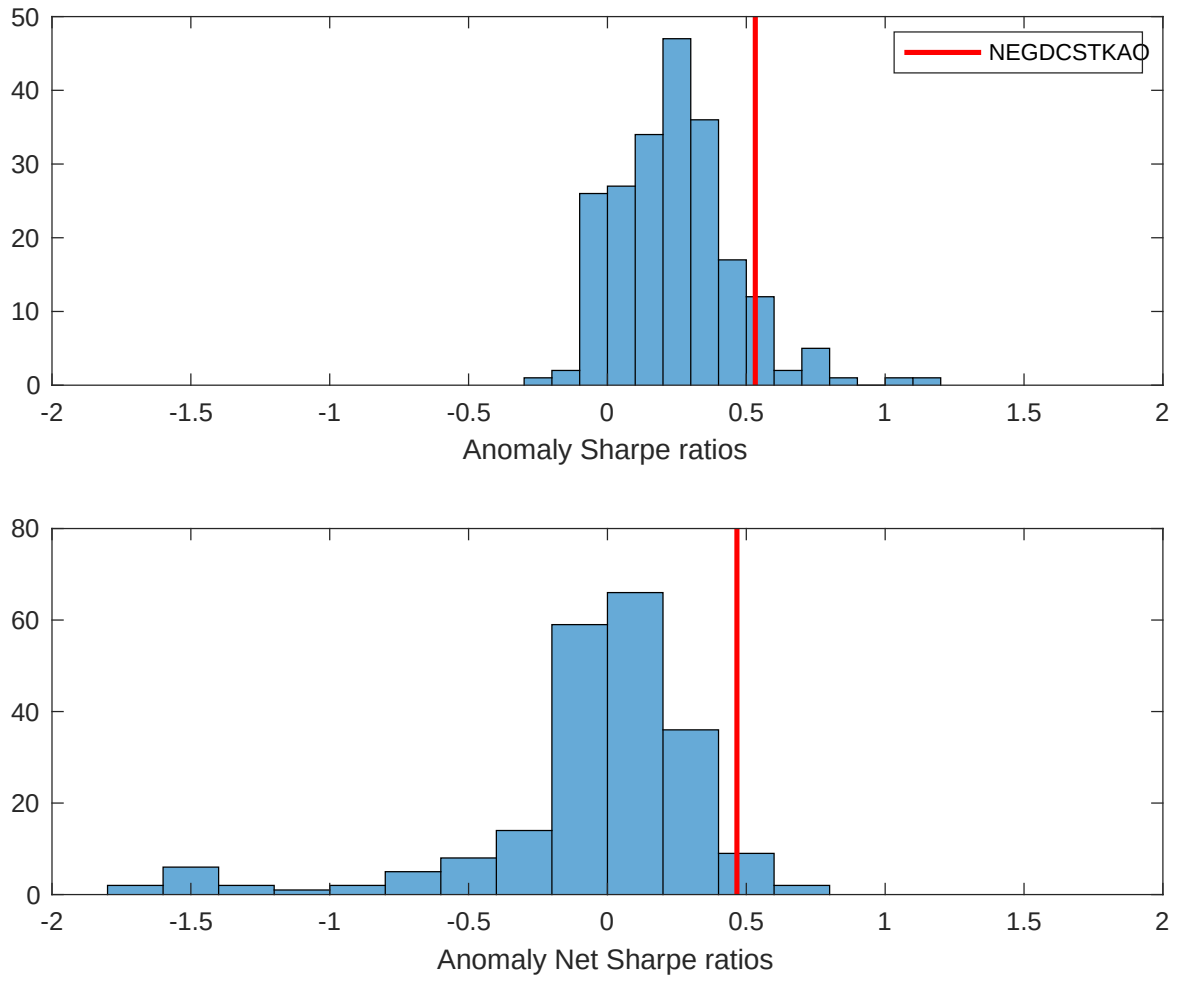


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EFI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

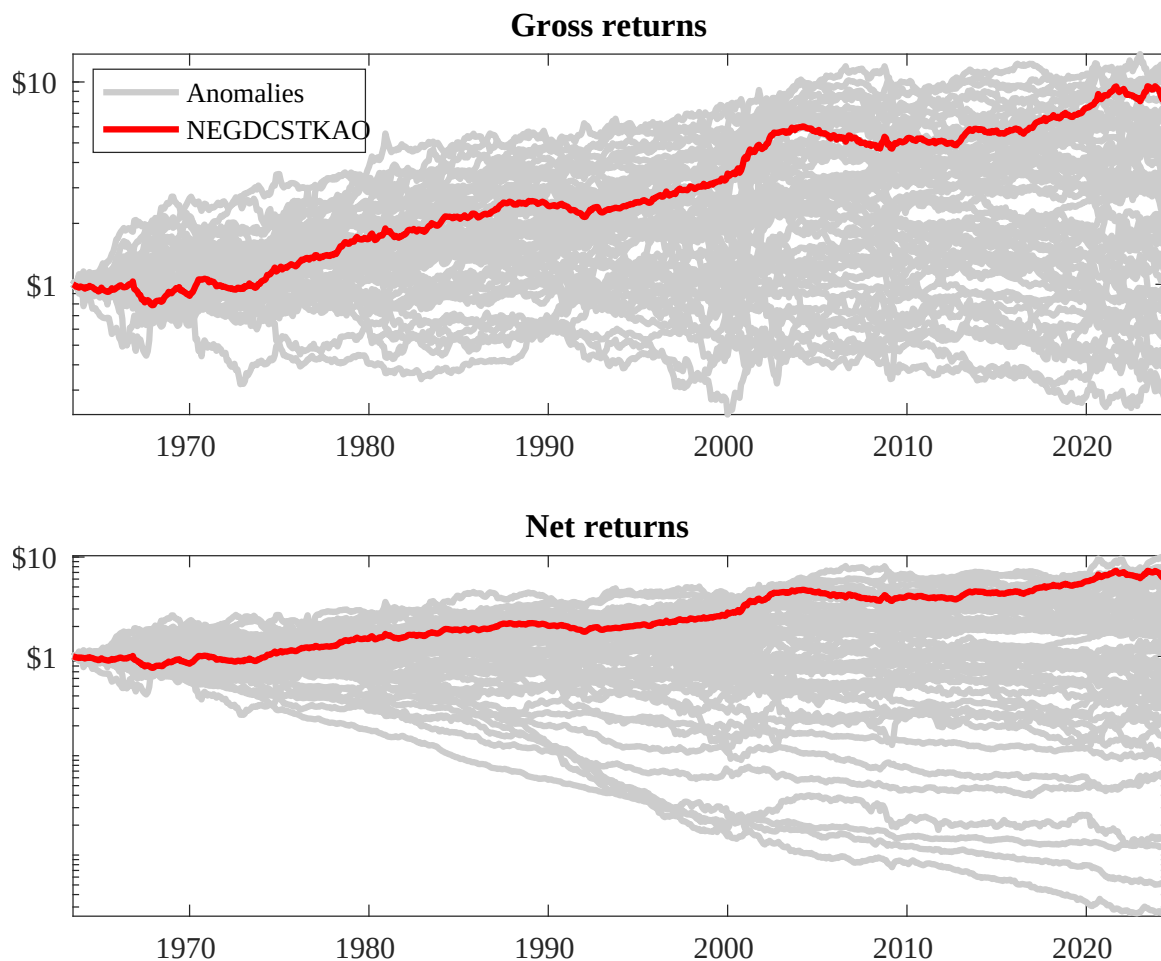
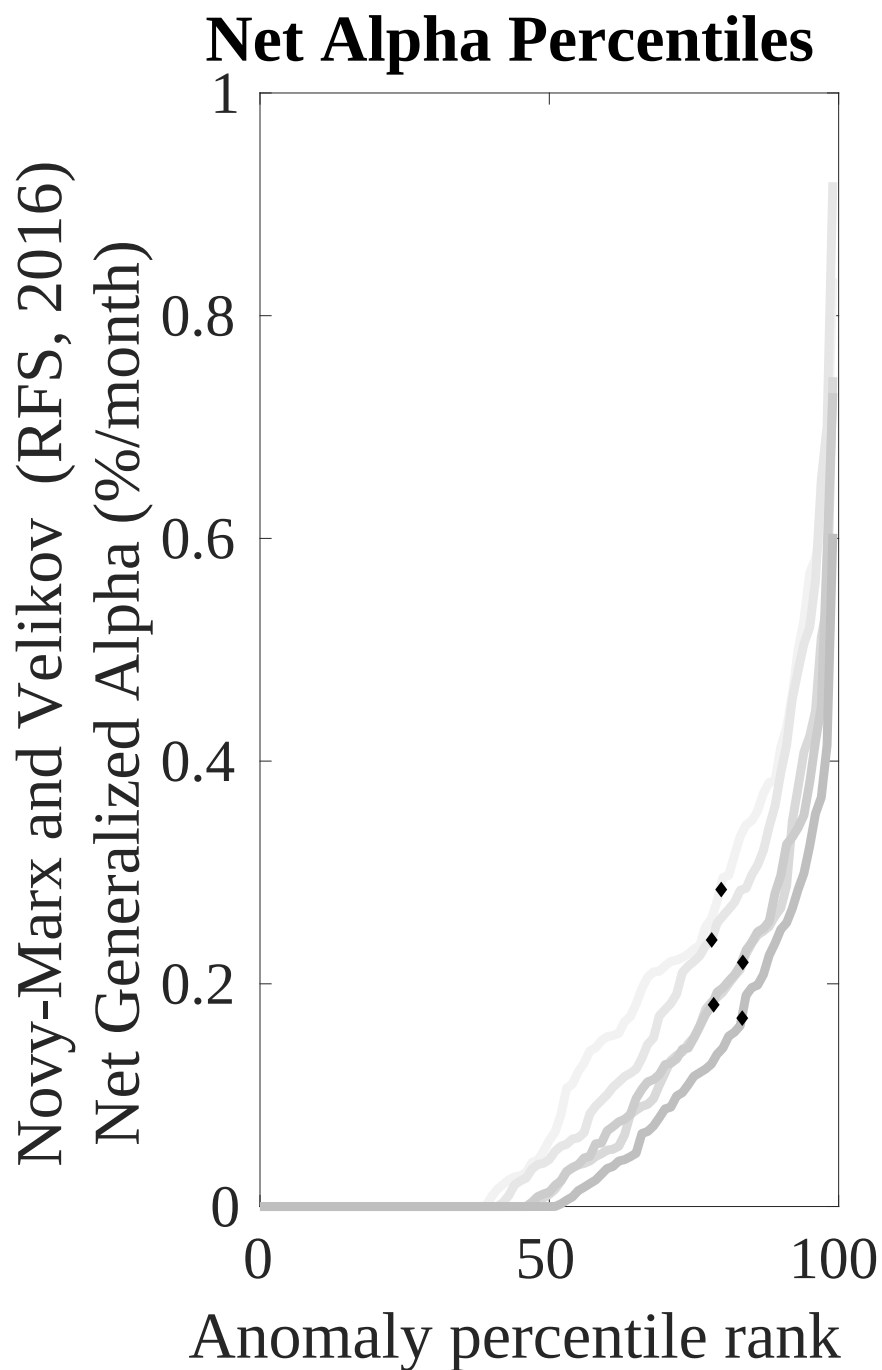
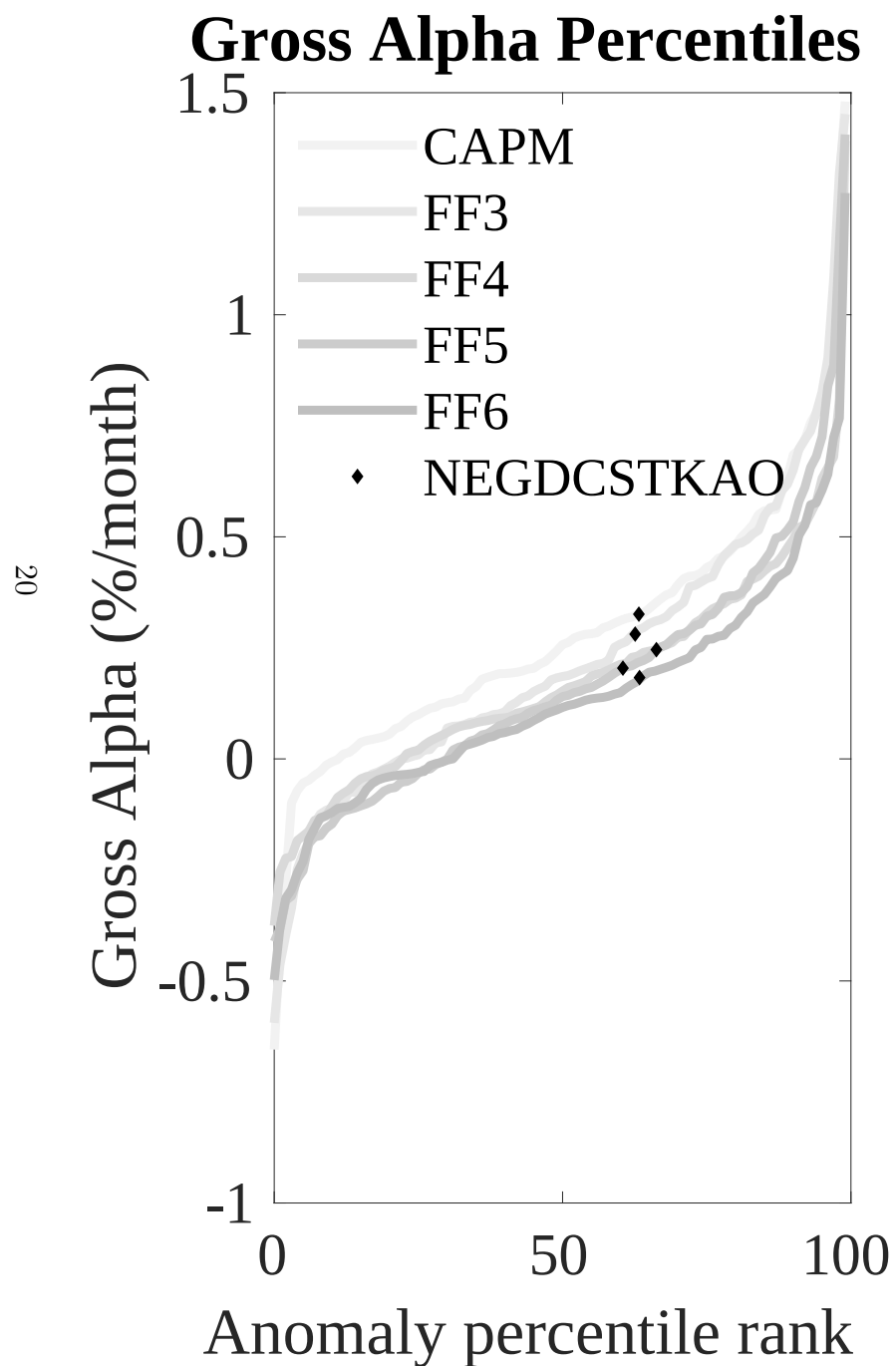


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EFI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



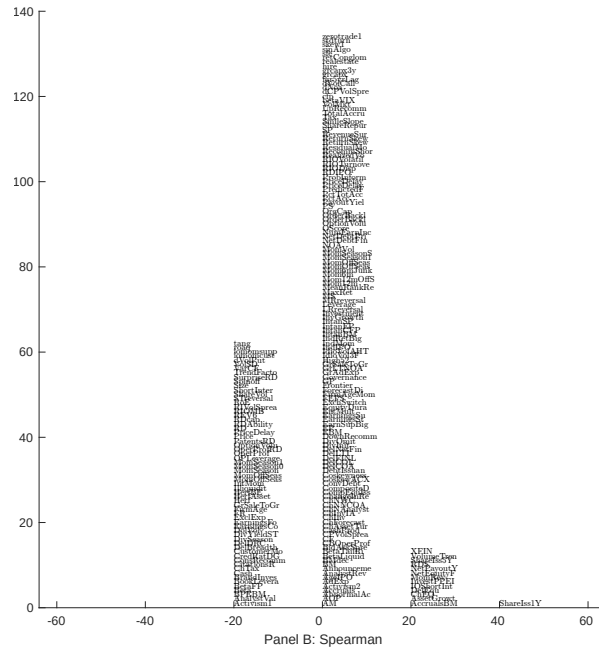
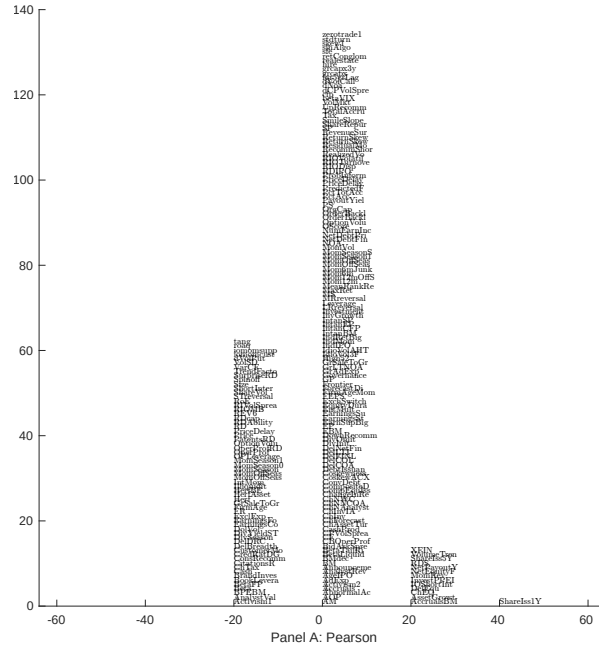


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with EFI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

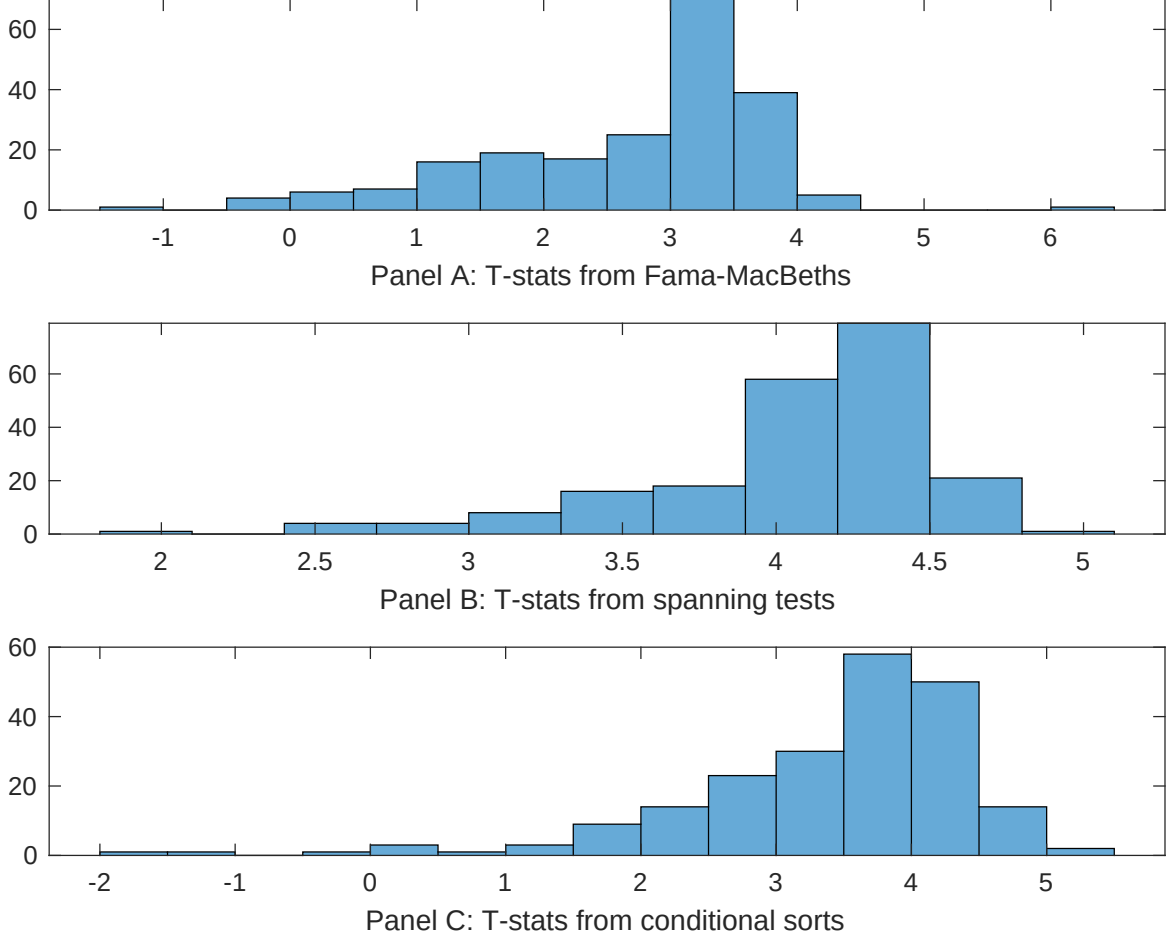


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EFI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EFI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EFI}EFI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EFI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EFI. Stocks are finally grouped into five EFI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EFI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EFI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EFI}EFI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Net Payout Yield, Share issuance (5 year), Growth in book equity, Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.05]	0.12 [5.43]	0.13 [6.43]	0.17 [6.98]	0.12 [5.72]	0.13 [5.35]	0.12 [4.59]
EFI	0.62 [2.43]	0.27 [0.91]	0.55 [2.23]	0.51 [1.81]	0.74 [2.55]	0.59 [2.05]	0.27 [0.85]
Anomaly 1	0.14 [6.54]						0.89 [4.06]
Anomaly 2		0.19 [1.66]					0.13 [2.74]
Anomaly 3			0.28 [4.97]				0.28 [6.16]
Anomaly 4				0.41 [3.51]			-0.16 [-1.11]
Anomaly 5					0.12 [3.31]		0.13 [2.13]
Anomaly 6						0.16 [2.60]	-0.14 [-1.77]
# months	715	715	715	720	720	630	622
$\bar{R}^2(\%)$	0	1	0	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EFI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EFI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Net Payout Yield, Share issuance (5 year), Growth in book equity, Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.16 [2.33]	0.21 [2.91]	0.11 [1.54]	0.16 [2.35]	0.18 [2.48]	0.22 [2.78]	0.17 [2.24]
Anomaly 1	27.62 [7.78]						10.02 [2.06]
Anomaly 2		15.85 [5.74]					10.48 [2.82]
Anomaly 3			28.29 [7.54]				19.07 [3.96]
Anomaly 4				26.48 [6.93]			26.78 [4.51]
Anomaly 5					12.11 [3.38]		-8.77 [-1.61]
Anomaly 6						8.82 [2.50]	-13.22 [-3.02]
mkt	2.77 [1.66]	4.55 [2.60]	3.36 [2.00]	2.95 [1.75]	1.86 [1.08]	3.50 [1.83]	3.65 [1.97]
smb	1.33 [0.55]	2.80 [1.12]	2.13 [0.88]	-0.60 [-0.25]	0.13 [0.05]	6.56 [2.15]	0.43 [0.15]
hml	2.21 [0.69]	-3.19 [-0.92]	5.21 [1.61]	-0.29 [-0.09]	1.36 [0.41]	2.92 [0.84]	0.41 [0.12]
rmw	-8.79 [-2.32]	-3.06 [-0.83]	-3.97 [-1.12]	7.41 [2.26]	7.27 [2.15]	6.05 [1.46]	-0.29 [-0.06]
cma	9.37 [1.80]	13.15 [2.47]	12.80 [2.53]	0.89 [0.15]	14.35 [2.36]	12.74 [2.24]	-13.88 [-2.10]
umd	1.93 [1.16]	4.45 [2.63]	1.34 [0.80]	2.73 [1.63]	3.50 [2.04]	3.93 [2.18]	2.69 [1.55]
# months	728	728	728	732	732	630	626
$\bar{R}^2(\%)$	16	13	16	15	10	8	18

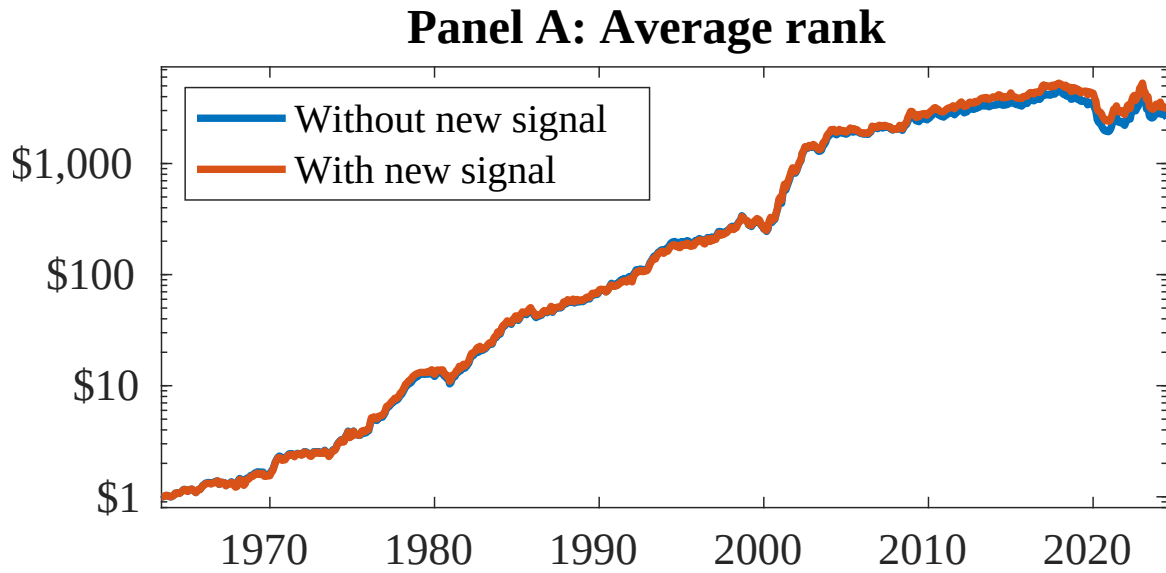


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EFI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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